

tube.xyz: an interactive structural atlas of the tubulin code

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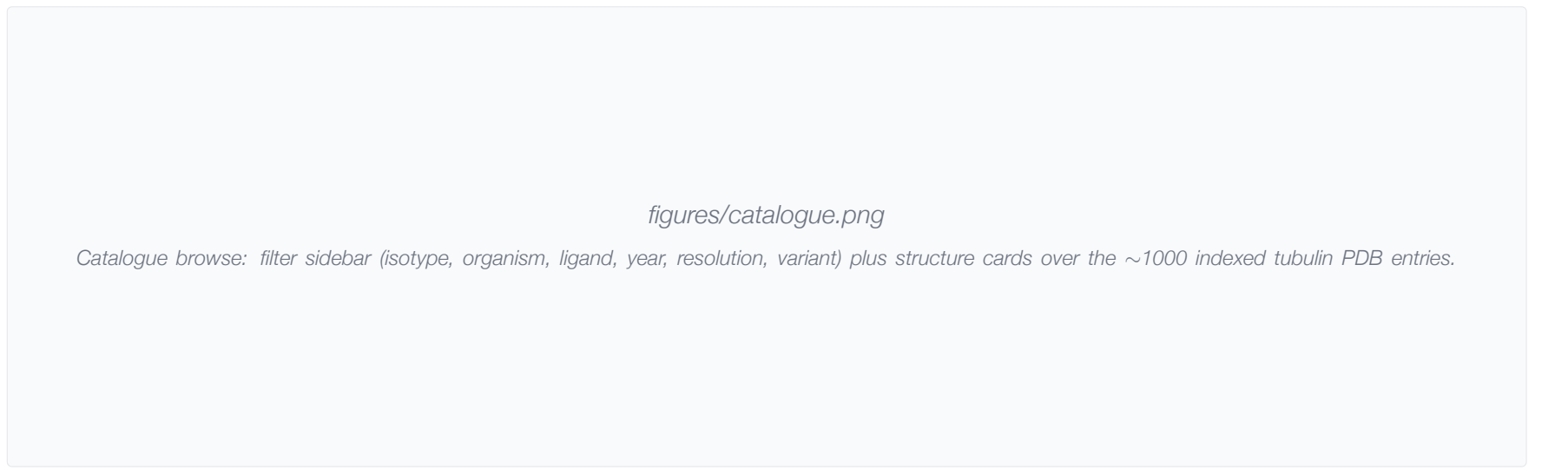
Motivation

The *tubulin code* — combinations of α/β isotypes and post-translational modifications — modulates microtubule composition and the interactions of tubulin with motors, MAPs and small molecules. The PDB now holds approximately one thousand tubulin-containing structures, capturing distinct conformational states and dozens of drug- and protein-binding pockets, while curated compendia such as the TubulinDB (Abbaali *et al.* 2023) catalogue thousands of disease-linked mutations and tubulin-specific PTMs. Despite the wealth on both sides, the structural and annotation layers live in separate resources and resist co-exploration.

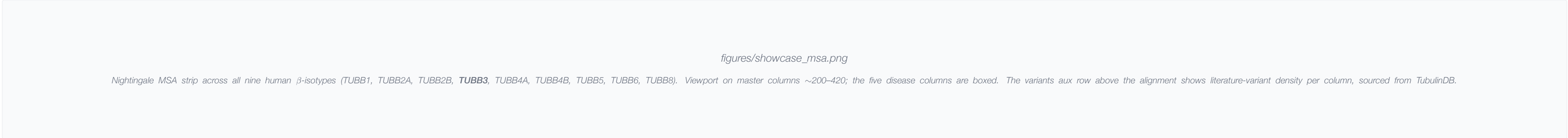
tube.xyz is a structural atlas that places every tubulin residue from every deposited structure into a single *coordinate system* per family (α , β , γ , δ , ε). Sequence variants, ligand-binding contacts, literature mutations and PTMs are programmatically lifted onto the common residue index. The substrate is deliberately **pluggable**: clinical variant catalogues, fragment screens and newly deposited structures join via a simple mapping with no schema changes. A 2D MSA is bidirectionally coupled to a Mol*-based 3D viewer; an LLM-driven chatbot translates plain-language questions about binding sites, cross-species residue comparisons, disease-linked mutations and PTMs into deep links into the integrated data.

Database overview

LAYER	SOURCE
Structures	RCSB PDB (IPR000217)
Tubulin families	HMM (α , β , γ , δ , ε)
MAP / enzyme families	HMM (kinesins, EBs, TTLs, ...)
tubHairline	UniProt (TUBA1A–TUBB8)
α/β isotypes	TubulinDB (Abbaali 2023)
Lit. mutations	TubulinDB + Janke 2020
Lit. modifications	Mol* → master coords
Binding sites	NCBI / ete3 lineages
Taxonomy	



1. Disease mutations in TUBB3 cluster at conserved positions of the structured face of β -tubulin

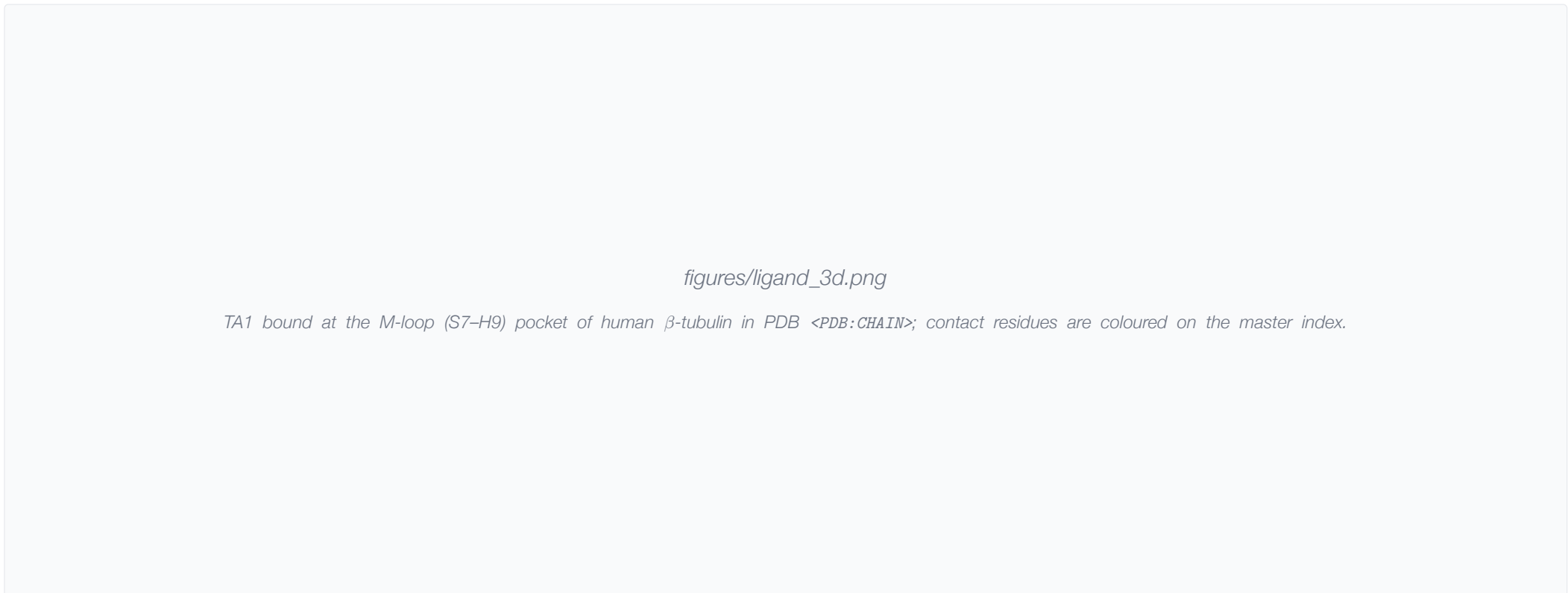


What the chatbot does. Asked “*which TUBB3 mutations cause CFEOM and where are they on the structure?*”, the chatbot resolves the five master positions, opens 6S8L:B in expert mode, paints the residues by phenotype, and aligns the nine human β -isotype references in the MSA panel so the conservation pattern is obvious at a glance.

What the figure shows. All five disease columns are identical across every human β -isotype — pathogenic precisely because they sit at residues the protein cannot afford to vary. The R→C/H substitutions at master 262 cause congenital fibrosis of the extraocular muscles (CFEOM); R→A at the same position is a biochemical kinesin-binding defect described in cell-free assays. Curation chain: Tischfield *et al.* 2010 *Cell* + Poirier *et al.* 2010 *Hum. Mol. Genet.* → TubulinDB (Abbaali 2023) → Neo4j via Universal Tubulin Numbering.

● CFEOM (extraocular fibrosis) ● cortical dysplasia / MCD ● biochemical / in-vitro phenotype

2. Taxane (TA1) contacts cluster on the M-loop of β -tubulin and project as an aux row onto the master alignment



What the chatbot does. “*Where does taxol bind in a human tubulin structure resolved at or below 3.5 Å?*” resolves to a faceted catalogue page filtered to human β -tubulin with TA1 bound, sorted by resolution; the top hit is opened in expert mode and the binding-site aux row is auto-toggled.

Why this matters. Taxane is the canonical microtubule-stabilising anti-cancer drug (Steinmetz & Prota 2018); its contacts span H6–H7, the M-loop (S7–H9) and S9–S10. Painting them as an aux row exposes the contact residues across every human β -isotype simultaneously — useful for predicting isotype selectivity and resistance-conferring positions.

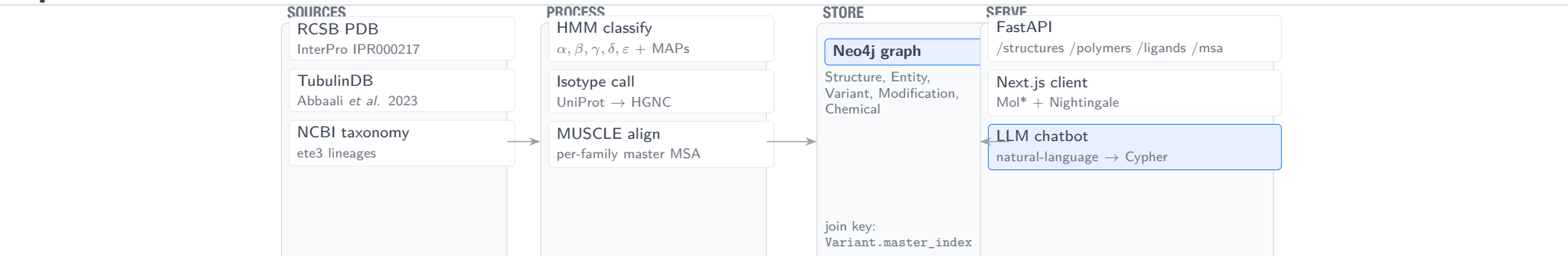
3. Phosphorylation tracks on TUBB3 reveal surface-accessible regulatory sites including β -S172



What the chatbot does. “*Show me phosphorylation sites on human beta-tubulin.*” stays on the central showcase chain (6S8L:B) and creates a custom aux row filtered to phosphorylation.

Cross-species transfer. The same paradigm carries human annotations onto sequences with no curated data: “... *in T.gondii tubulin* — *compare sequences and highlight phosphorylation sites.*” aligns the host chain onto the human master and overlays the human-derived PTM track on top of the *T.gondii* sequence in the MSA panel.

Pipeline



References & links

Key refs. [1] Abbaali *et al.* (2023) *PLoS ONE* **18**: e0295279 (TubulinDB). [2] Janke & Magiera (2020) *Nat. Rev. Mol. Cell Biol.* **21**, 307. [3] Steinmetz & Prota (2018) *Trends Cell Biol.* **28**, 776. [4] Tischfield *et al.* (2010) *Cell* **140**, 74 (TUBB3 disease mutations).

ACKNOWLEDGEMENTS

Funding / institutional affiliations TBC. Live at **tube.xyz** · code at github.com/rtviii/tubulinxyz.



TUBE.XYZ
the live atlas.
GITHUB
backend + frontend repos.